FIELDMARK INSTRUMENTS

AquaSense 300

Capacitive soil moisture and temperature sensor

Overview

The AquaSense 300 measures volumetric water content and soil temperature at the root zone of field crops, orchards and school gardens. Its sealed probe has no exposed metal electrodes, so readings stay stable for several seasons in acidic or saline soils.

The sensor is installed by pushing it into a pilot hole made with the supplied auger. A single cable carries both power and data, which keeps installation simple for irrigation contractors and extension officers.

Features

- Measures moisture at three depths along one probe
- Built-in temperature compensation from −10 to 55 °C
- SDI-12 and Modbus RTU outputs on the same cable
- IP68 housing rated for permanent burial
- Factory calibration for mineral soils, with a user table for peat

Tip: Install the probe at least 30 cm from the nearest drip emitter. Readings taken closer than this mostly show the wetting pattern of the emitter rather than the moisture of the field.

Typical applications

Scheduling irrigation for vegetables and fruit trees, research trials on water use, and teaching soil science with live data. Several sensors can share one logger, and the logger can send readings to a phone every fifteen minutes.

Specifications

Parameter	Value
Measurement	
Moisture range	0 to 60% VWC
Accuracy	±2% VWC after calibration
Depths	10, 20 and 30 cm
Temperature accuracy	±0.5 °C
Electrical	
Supply voltage	6 to 16 V DC
Current	12 mA active, 30 µA sleep
Outputs	SDI-12 v1.4, Modbus RTU (RS-485)
Mechanical	
Probe length	380 mm
Housing	Glass-filled polyamide, IP68
Operating range	−20 to 60 °C

Ordering information

Order code	Cable	Price (EUR)
AS300-05	5 m	214.00
AS300-10	10 m	229.00
AS300-25	25 m	262.00

Prices exclude VAT and delivery. Volume discounts are available for orders of ten or more sensors.

Specifications are subject to change without notice.